

Frederick Tombs

fredtombs.com



fgitombs



gigeresk

fredtombs@gmail.com | +1 (603) 503-0488 | English and French | Boston, Massachusetts

Hardware/software engineer specializing in FPGA architecture, electronic design automation tool development, and silicon tapeout. I am comfortable at every level of the stack, from Python, to C++, to the transistor level.

WORK EXPERIENCE

Zero ASIC

Cambridge, Massachusetts, U.S.A.

Senior Software Engineer

April 2025 – Present

Zero ASIC is building a composable chiplet platform centered around FPGAs, and the tools to make it happen.

- Tapeout of an FPGA SoC chiplet on the GlobalFoundries FinFET 12LP+ process with Synopsys Fusion Compiler after a six-month schedule pull-in. 50K LUT heterogeneous architecture on 4x4mm.
 - Designed a process-agnostic pin planning tool for tileable FPGAs.
 - Timing and power sign-off flows spanning all PVT corners via tile model reuse; caught a critical hold violation and 3 RTL bugs pre-tapeout that would have required a respin or CAD workarounds.
 - Built a liberty file generator for FPGA core integration at the SoC top-level (Python, OpenSTA, SiliconCompiler); characterized using Synopsys Library Compiler.
 - RTL-to-Gate formal verification with Synopsys Formality.
- **FPGA Architect:** Architected performance improvements for an FPGA generation tool; architectural decisions drove a 10x speedup and 50% memory reduction, unblocking design and runtime scalability for tapeout.
 - Built a timing model extraction and annotation flow for embedded FPGAs, enabling timing closure.
- **Logik:** Led development for an open-source FPGA EDA toolchain.
 - Extended toolchain with VHDL support, a pin assignment tool, and CLI.
 - **ABC** (open-source logic synthesis/verification): Identified and patched an 18-year-old pointer-truncation bug causing intermittent failures; built a regression framework to prevent recurrence (C).
 - **VTR:** Reduced runtime by 9% and memory usage by 8% by replacing glibc malloc with jemalloc.
- **In-development ASIC synthesis tool:** Proposed and led development of a cloud-hosted EDA frontend as the primary distribution strategy for a novel synthesis tool; built and shipped the full stack with a team of 2.
- **wildebeest** (FPGA synthesis tool): Implemented hard-block tech mapping passes to reduce design area (C++).

Efinix, Inc.

Toronto, Ontario, Canada

Senior Software Engineer

Feb. 2021 – Feb. 2025

Efinix develops low-power FPGAs competing in a market dominated by AMD (Xilinx) and Altera.

- **Detailed placement:** Led development of the detailed placement optimization engine (C++17/20)
 - Tripled runtime performance on large designs through algorithmic tuning and hotspot parallelization.
 - Integrated reinforcement-learning move selection to accelerate convergence and improve solution quality, enabling 1.5x speedup at no loss in quality.
 - Implemented power-aware placement optimizations, resulting in >10% dynamic power savings for customer designs.
- **Placement legalization:** Owned the placement legalization algorithm (dynamic-programming heuristic)
 - Parallelized a performance hotspot using Intel TBB, extended support for new FPGA architectures, and resolved high-priority customer issues.
- **Primitive modeling framework:** Spearheaded a transition towards a data-centric, cache-aware device modeling framework (C++, Python, protobuf), improving runtime efficiency and accelerating feature development.
 - Designed a primitive/atom-based netlist using Verific, enabling physical synthesis (5-8% Fmax).
- Built and operationalized a hardware lab for EDA-silicon correlation. Secured executive buy-in and led procurement. Enabled quick iteration loop for major timing and power improvements.

EDUCATION

B.A.Sc, Honours Computer Engineering (Co-op), **University of Waterloo**

PUBLICATIONS

F. Tombs, A. Mellat and N. Kapre, “Mocarabe: High-Performance Time-Multiplexed Overlays for FPGAs,” *2021 IEEE 29th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM)*

- Hardware architecture and compiler to generate overlays for Xilinx Ultrascale+ FPGAs from C kernels.
- Stochastic-search placement, space-time PathFinder router, ILP-based router (Python, Gurobi, SCIP).
- 22% acceptance rate. Key results: outperform Vivado HLS clock frequency up to 9.2x (650 MHz).